

KamillLabs Edu AI

API Documentation

Frontend Integration Guide
Version 3.1 | Multi-Provider Architecture
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1. Overview & Architecture

1.1 System Overview

KamillLabs Edu AI is a comprehensive AI-powered educational platform built on Laravel 12 that provides three core features for Indonesian schools: Learning Recommendations, Teaching Material Generation, and RPP (Lesson Plan) Generation.

The system supports multiple AI providers (Anthropic Claude and Google Gemini) with dual-mode operation supporting both synchronous and asynchronous processing via Laravel queues. Providers can be switched via environment configuration without any code changes.

1.2 Architecture Patterns

The codebase follows a clean, layered architecture with single-responsibility layers:

Layer	Location	Responsibility
Controllers	Http/Controllers/Api/	Thin orchestrators that delegate to other layers
Form Requests	Http/Requests/{Feature}/	Input validation and rate limit pre-checks
Actions	Actions/{Feature}/	Business logic for writes and mutations
Repositories	Repositories/	Database read abstractions
API Resources	Http/Resources/	Response formatting with success wrapper
Services	Services/AI/	AI integration and generation logic
Models	Models/	Eloquent models with UUID primary keys
Enums	Enums/	PHP 8.4 string-backed enums

1.3 Authentication & Middleware

All API routes require the following middleware stack:

Middleware	Purpose
auth:sanctum	Bearer token authentication via Laravel Sanctum
role.teacher	Ensures the authenticated user has a teacher role
teacher.owner	Verifies the teacher owns the requested resource

Header: Include with every request:

```
Authorization: Bearer {your_sanctum_token}
```

1.4 Base URL & Response Format

```
Base URL: https://your-domain.com/api
```

Success response:

```
{
  "success": true,
  "data": "...",
  "message": "Optional message"
}
```

Error response:

```
{
  "success": false,
  "message": "Error description in Bahasa Indonesia"
}
```

Code	Meaning
200	OK - Request successful
201	Created - Resource created successfully
202	Accepted - Job queued for async processing (includes poll_url)
404	Not Found - Resource does not exist
422	Validation Error - Invalid input or business rule violation
429	Rate Limited - Daily/monthly generation limit exceeded

1.5 Dual-Mode Architecture (Sync vs Async)

The system supports two processing modes based on the queue driver configuration:

Mode	Queue Driver	Behavior	Response
Sync	sync	Returns result immediately	200 / 201
Async	redis / database	Dispatches to queue, returns job ID	202

Async response example (202):

```
{
  "success": true,
  "message": "RPP sedang di-generate. Silakan cek status secara berkala.",
  "job_id": "uuid-of-job",
  "poll_url": "/api/ai-jobs/{job_id}"
}
```

When receiving a 202 response, the frontend should poll the job status endpoint every 5 seconds until the job is completed or failed.

1.6 Rate Limiting

Each AI feature has daily and monthly generation limits per teacher. Limits are checked in two places: Form Request validation (early rejection) and Service layer (safety net for queue jobs).

Feature	Daily	Monthly	Cooldown
Recommendations	10 / teacher	100 / teacher	5 minutes
Teaching Materials	20 / teacher	500 / teacher	None
RPP (Lesson Plans)	15 / teacher	200 / teacher	None

Rate limit error (429):

```
{
  "success": false,
  "message": "Batas pembuatan RPP AI harian/bulanan telah tercapai."
}
```

1.7 AI Provider Switching

The system supports multiple AI providers that can be switched via the AI_PROVIDER environment variable. All feature services use the AIServiceInterface contract, making the switch transparent to business logic.

Configuration

```
# .env file
AI_PROVIDER=anthropic    # or "gemini"

# Anthropic (Claude)
ANTHROPIC_API_KEY=sk-ant-...
```

```
# Google Gemini
GEMINI_API_KEY=your-google-ai-studio-key
```

Provider Registration

The AIServiceProvider uses a match expression to resolve the correct implementation based on the configured provider:

Provider	ENV Value	Service Class	API Key Transport
Anthropic Claude	anthropic (default)	ClaudeAIService	Header: x-api-key
Google Gemini	gemini	GeminiAIService	Query param: ?key=

Shared Interface

Both providers implement the same AIServiceInterface contract with these methods:

Method	Description
sendMessage()	Send a prompt and get text response with token counts
sendMessageJson()	Send a prompt and get parsed JSON response
defaultModel()	Returns the configured default (complex) model
fastModel()	Returns the configured fast (lightweight) model
totalTokens()	Calculate total tokens from a response

Provider switching is transparent to the frontend. The API endpoints, request parameters, and response format remain identical regardless of which AI provider is active.

1.8 AI Models Comparison

Anthropic Claude Models

Model	ID	Used For
Claude Sonnet 4.5	claude-sonnet-4-5-20250929	Complex tasks: RPP generation, recommendations, materials
Claude Haiku 4.5	claude-haiku-4-5-20251001	Lightweight: quiz/reference regen, simple RPP fields

Google Gemini Models

Model	ID	Used For
Gemini 2.5 Flash	gemini-2.5-flash	Complex tasks (default model)
Gemini 2.5 Flash Lite	gemini-2.5-flash-lite	Lightweight tasks (fast model)

Feature Differences

Feature	Claude (Anthropic)	Gemini (Google)
Prompt Caching	Supported (up to 90% cost reduction)	Not available
JSON Mode	Via prompt instruction	Native responseType
Batch API	Supported (sendBatch)	Not implemented
Retry Logic	2 retries on server errors	2 retries on 5xx / connection errors
System Prompt	Native system parameter	Via systemInstruction field
Stop Reasons	end_turn, max_tokens	Normalized: STOP, MAX_TOKENS, SAFETY, RECITATION

2. AI Job Status API

When the system operates in async mode, AI generation requests are dispatched to a background queue. This API allows the frontend to poll for job progress and retrieve results.

2.1 Job Lifecycle

Status	Description	Frontend Action
pending	Job created, waiting in queue	Show spinner, continue polling
processing	Worker picked up the job	Show progress indicator, continue polling
completed	Job finished successfully	Stop polling, display result data
failed	Job failed after max retries	Stop polling, show error message

Configuration: Max 2 retries, 30-second backoff, 660-second timeout. Jobs use WithoutOverlapping middleware keyed by teacher_id:feature.

2.2 Get Job Status

GET /api/ai-jobs/{id}

Returns the status and result of a specific AI generation job.

Completed Job Response

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "feature": "rpp",
    "action": "generate",
    "status": "completed",
    "attempts": 1,
    "started_at": "2026-03-02T10:00:00.000000Z",
    "completed_at": "2026-03-02T10:00:45.000000Z",
    "result": {
      "id": "uuid",
      "title": "RPP Matematika",
      "status": "draft",
      "...": "full model"
    },
    "ai_model_used": "claude-sonnet-4-5-20250929",
    "ai_tokens_used": 3500
  }
}
```

```
}
}
```

Failed Job Response

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "feature": "rpp",
    "action": "generate",
    "status": "failed",
    "attempts": 2,
    "error_message": "AI service temporarily unavailable",
    "started_at": "2026-03-02T10:00:00.000000Z",
    "completed_at": "2026-03-02T10:01:00.000000Z"
  }
}
```

2.3 List Jobs

GET /api/ai-jobs

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
feature	string	No	Filter: recommendations, teaching_material, rpp
status	string	No	Filter: pending, processing, completed, failed
per_page	integer	No	Items per page (1-50, default 15)

2.4 Polling Strategy

Recommended frontend polling implementation:

```
async function pollJob(jobId) {
  const maxAttempts = 60; // 5 minutes max
  for (let i = 0; i < maxAttempts; i++) {
    const res = await fetch(`/api/ai-jobs/${jobId}`);
    const { data } = await res.json();

    if (data.status === "completed") return data.result;
    if (data.status === "failed") throw new Error(data.error_message);

    await new Promise(r => setTimeout(r, 5000));
  }
}
```

```
    }  
    throw new Error("Job timed out");  
}
```

3. Learning Recommendations API

The AI Learning Recommendations system analyzes student performance data (grades, assessments, attendance) to generate personalized learning recommendations with associated materials.

3.1 Data Model

Field	Type	Description
id	uuid	Primary key
school_id	uuid	FK to schools
student_id	uuid	FK to students
class_id	uuid	FK to classes
subject_id	uuid	FK to subject_schools
teacher_id	uuid	FK to teachers
type	enum	remedial enrichment strategy
category	enum	weak_topic declining_trend attendance_warning enrichment_opportunity study_strategy
trigger_source	enum	on_demand post_exam weekly_review attendance_alert
title	string	Recommendation title
description	text/html	Detailed description
ai_reasoning	text	AI explanation for this recommendation
priority	enum	high medium low
performance_data	json	Student performance metrics used by AI
status	enum	pending in_progress completed dismissed
completed_at	datetime	When marked completed (auto-managed)

3.2 Generate for Class

POST /api/recommendations/generate

Generate AI recommendations for all students in a class. Analyzes each student's performance and produces personalized recommendations with learning materials.

Request Body

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
class_id	uuid	Yes	Class UUID
subject_id	uuid	Yes	Subject UUID (subject_schools table)
trigger_source	string	No	on_demand post_exam weekly_review attendance_alert

Sync Response (201)

```
{
  "success": true,
  "message": "Rekomendasi berhasil dibuat untuk 25 siswa.",
  "data": [
    {
      "id": "uuid",
      "student_id": "uuid",
      "type": "remedial",
      "category": "weak_topic",
      "priority": "high",
      "title": "Penguatan Materi Aljabar Linear",
      "description": "<p>Siswa menunjukkan kesulitan...</p>",
      "ai_reasoning": "Berdasarkan analisis nilai...",
      "status": "pending",
      "materials": [
        {
          "id": "uuid",
          "title": "Latihan Soal Aljabar",
          "type": "exercise",
          "content": "<p>...</p>"
        }
      ]
    }
  ],
  "count": 25
}
```

Async Response (202)

```
{
  "success": true,
  "message": "Rekomendasi sedang di-generate.",
  "job_id": "uuid",
  "poll_url": "/api/ai-jobs/{job_id}"
}
```

3.3 Generate for Student

POST /api/recommendations/generate-student

Generate a recommendation for a single specific student.

Request Body

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
student_id	uuid	Yes	Student UUID
class_id	uuid	Yes	Class UUID
subject_id	uuid	Yes	Subject UUID

3.4 List Recommendations

GET /api/recommendations

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
class_id	uuid	No	Filter by class
subject_id	uuid	No	Filter by subject
student_id	uuid	No	Filter by student
type	string	No	remedial enrichment strategy
category	string	No	weak_topic declining_trend etc.
priority	string	No	high medium low
status	string	No	pending (default) in_progress completed dismissed
per_page	integer	No	Items per page (1-50, default 15)

Response is paginated with standard Laravel pagination (data, links, meta).

3.5 Show Detail

GET /api/recommendations/{id}

Returns recommendation with student, subject, materials, teacher, and chapter relations loaded. Uses RecommendationResource for formatting.

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "student_id": "uuid",
    "type": "remedial",
    "category": "weak_topic",
    "priority": "high",
    "title": "Penguatan Materi Aljabar",
    "description": "<p>Deskripsi lengkap...</p>",
    "ai_reasoning": "Berdasarkan analisis performa...",
    "performance_data": {
      "average_score": 45,
      "assessment_count": 5,
      "trend": "declining"
    },
    "status": "pending",
    "materials": [
      {
        "id": "uuid",
        "title": "Latihan Soal",
        "type": "exercise"
      }
    ],
    "created_at": "2026-03-02T10:00:00.000000Z"
  }
}
```

3.6 Update Status

PATCH /api/recommendations/{id}/status

Update recommendation status. The UpdateRecommendationStatusAction automatically manages timestamps: sets completed_at when marking completed, clears it when reverting.

Request Body

Parameter	Type	Required	Description
status	string	Yes	pending in_progress completed dismissed

3.7 Class Summary

GET /api/recommendations/class/{class_id}/summary

Aggregated summary of recommendations grouped by category, priority, and status. Useful for dashboard widgets.

```
{
  "success": true,
  "data": {
    "by_category": {
      "weak_topic": 12,
      "declining_trend": 5,
      "attendance_warning": 3,
      "enrichment_opportunity": 8,
      "study_strategy": 4
    },
    "by_priority": {
      "high": 10,
      "medium": 15,
      "low": 7
    },
    "by_status": {
      "pending": 20,
      "in_progress": 5,
      "completed": 3,
      "dismissed": 4
    }
  }
}
```

4. Teaching Material Generator API

Generates comprehensive teaching content for a chapter including main material, quiz questions, and references. Results are cached per teacher+chapter combination.

4.1 Data Model

Generated Material

Field	Type	Description
id	uuid	Primary key
teacher_id / school_id	uuid	Foreign keys
subject_id	uuid	FK to subject_schools
chapter_id	uuid	FK to chapters
sub_chapter_id	uuid	FK to sub_chapters (nullable)
material_content	text/html	Main teaching content
material_content_generated_at	datetime	Generation timestamp
ai_model_used	string	Claude model used
ai_tokens_used	integer	Total tokens consumed

Generated Quiz (HasMany)

Field	Type	Description
question	text	Quiz question text
question_type	string	multiple_choice true_false short_answer essay
options	json	Answer options array (for multiple choice)
correct_answer	string	The correct answer
explanation	text	Explanation of the answer
difficulty	string	easy medium hard
generation_batch	integer	Batch number for tracking regenerations

Generated Reference (HasMany)

Field	Type	Description
title	string	Reference title
content	text/html	Reference content

Field	Type	Description
type	string	concept_deep_dive real_world_example common_misconception teaching_tip
generation_batch	integer	Batch number

4.2 Generate Material

POST /api/generated-materials/generate

Generates teaching material, quiz questions, and references in a single call. Returns cached result if material already exists for this teacher + chapter combination.

Request Body

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
subject_id	uuid	Yes	Subject UUID (subject_schools)
chapter_id	uuid	Yes	Chapter UUID
sub_chapter_id	uuid	No	Sub-chapter UUID

Response (200)

```
{
  "success": true,
  "data": {
    "id": "uuid",
    "material_content": "<h2>Materi Aljabar</h2><p>...</p>",
    "quizzes": [
      {
        "id": "uuid",
        "question": "Jika  $2x + 3 = 11$ , berapakah nilai  $x$ ?",
        "question_type": "multiple_choice",
        "options": [
          "x = 3",
          "x = 4",
          "x = 5",
          "x = 6"
        ],
        "correct_answer": "x = 4",
        "explanation": " $2x = 8$ ,  $x = 4$ ",
        "difficulty": "easy",
        "generation_batch": 1
      }
    ]
  }
}
```

```

    "references": [
      {
        "id": "uuid",
        "title": "Konsep Dasar Aljabar",
        "content": "<p>Penjelasan mendalam...</p>",
        "type": "concept_deep_dive",
        "generation_batch": 1
      }
    ]
  },
  "cached": false
}

```

4.3 Show Material

GET /api/generated-materials/{id}

Returns material with quizzes and references. Uses GeneratedMaterialResource with regeneration status metadata.

```

{
  "success": true,
  "data": {
    "id": "uuid",
    "material_content": "...",
    "quizzes": [
      "..."
    ],
    "references": [
      "..."
    ]
  },
  "regen_status": {
    "quiz": {
      "remaining": 2,
      "can_regenerate": true,
      "count": 1,
      "max": 3
    },
    "reference": {
      "remaining": 1,
      "can_regenerate": true,
      "count": 1,
      "max": 2
    }
  }
}

```

4.4 List Materials

GET /api/generated-materials

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
subject_id	uuid	No	Filter by subject
chapter_id	uuid	No	Filter by chapter
per_page	integer	No	Items per page (1-50, default 15)

4.5 Regenerate Quiz

POST /api/generated-materials/{id}/regenerate-quiz

ADDS new questions to the existing set (does not replace). Uses Claude Haiku. Max 3 regenerations, 5-minute cooldown.

```
{
  "success": true,
  "data": [
    {
      "id": "uuid",
      "question": "Tentukan himpunan penyelesaian...",
      "question_type": "short_answer",
      "correct_answer": "x > 4",
      "difficulty": "medium",
      "generation_batch": 2
    }
  ],
  "remaining": 1,
  "can_regenerate": true
}
```

4.6 Regenerate References

POST /api/generated-materials/{id}/regenerate-reference

REPLACES all existing references (deletes old ones first). Uses Claude Haiku. Max 2 regenerations, 5-minute cooldown.

```
{
```

```

"success": true,
"data": [
  {
    "id": "uuid",
    "title": "Aplikasi Aljabar",
    "content": "<p>...</p>",
    "type": "real_world_example",
    "generation_batch": 2
  }
],
"remaining": 0,
"can_regenerate": false
}

```

4.7 Check Cache

GET /api/generated-materials/check-cache

Check if cached material exists before triggering generation.

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
chapter_id	uuid	Yes	Chapter UUID
sub_chapter_id	uuid	No	Sub-chapter UUID

```

{
  "cached": true,
  "material_id": "uuid-of-cached-material"
}

```

5. RPP (Lesson Plan) Generator API

Creates comprehensive Rencana Pelaksanaan Pembelajaran following Indonesian curriculum standards. Supports a full workflow: generate, edit, regenerate fields, submit for review, and admin approval/revision.

5.1 Data Model

Each RPP contains 10 AI-generated content fields:

Field	Description
core_competence	Kompetensi Inti (KI)
basic_competence	Kompetensi Dasar (KD)
indicator	Indikator Pencapaian Kompetensi
learning_objective	Tujuan Pembelajaran
main_material	Materi Pokok
learning_method	Metode Pembelajaran
media_tools	Media dan Alat Pembelajaran
learning_source	Sumber Belajar
learning_activities	Kegiatan Pembelajaran (Complex - uses Sonnet)
assessment	Penilaian (Complex - uses Sonnet)

Additional metadata fields: title, semester, academic_year, status, note_admin, ai_generated, ai_model_used, ai_tokens_used, submitted_at, approved_at, approved_by, export_count.

5.2 Status Workflow

Status	Description	Next Status	Who
draft	Initial state after generation	submitted	Teacher (submit)
submitted	Awaiting admin review	approved / revision	Admin
approved	Final approved state	None (terminal)	N/A
revision	Returned for edits	submitted	Teacher (submit)

Only draft and revision RPPs are editable. Only editable RPPs can be submitted. Only approved

RPPs can be exported to DOCX.

5.3 Generate RPP

POST /api/lesson-plans/generate

Generate a complete RPP via AI. All 10 content fields are populated. Created in draft status.

Request Body

Parameter	Type	Required	Description
teacher_id	uuid	Yes	Teacher UUID
subject_id	uuid	Yes	Subject UUID (subject_schools)
class_id	uuid	Yes	Class UUID
chapter_id	uuid	Yes	Chapter UUID
sub_chapter_id	uuid	No	Sub-chapter UUID
time_allocation	string	No	e.g. "2 x 45 menit" (max 50 chars)

Sync Response (201)

```
{
  "success": true,
  "message": "RPP berhasil di-generate oleh AI.",
  "data": {
    "id": "uuid",
    "title": "RPP Matematika - Aljabar Linear Kelas X",
    "core_competence": "<p>Menghayati dan mengamalkan...</p>",
    "basic_competence": "<p>3.1 Mendeskripsikan konsep...</p>",
    "indicator": "<ol><li>Mengidentifikasi variabel...</li></ol>",
    "learning_objective": "<ol><li>Siswa dapat menjelaskan...</li></ol>",
    "main_material": "<h3>Aljabar Linear</h3><ul><li>Definisi...</li></ul>",
    "learning_method": "<p>Pendekatan: Scientific Learning</p>",
    "media_tools": "<ul><li>Papan tulis</li><li>Proyektor</li></ul>",
    "learning_source": "<ul><li>Buku Paket Kelas X</li></ul>",
    "learning_activities": "<h3>Pendahuluan (10 menit)</h3><p>...</p>",
    "assessment": "<h3>Penilaian Pengetahuan</h3><p>...</p>",
    "status": "draft",
    "ai_generated": true,
    "ai_model_used": "claude-sonnet-4-5-20250929",
    "ai_tokens_used": 3200
  },
  "regen_limits": {
    "core_competence": {
      "used": 0,

```

```

    "max": 2,
    "remaining": 2
  },
  "basic_competence": {
    "used": 0,
    "max": 2,
    "remaining": 2
  },
  "...": "all 10 fields"
}

```

5.4 Show RPP

GET /api/lesson-plans/{id}

Returns RPP detail with relations, regeneration limits, and status flags. Uses LessonPlanResource.

```

{
  "success": true,
  "data": {
    "...": "full lesson plan with loaded relations"
  },
  "regen_limits": {
    "core_competence": {
      "used": 1,
      "max": 2,
      "remaining": 1
    },
    "...": "all fields"
  },
  "is_editable": true,
  "can_submit": true,
  "can_export": false
}

```

5.5 Update Fields (Auto-Save)

PATCH /api/lesson-plans/{id}

Update one or more RPP fields. Only works when editable (draft/revision). Uses UpdateLessonPlanFieldsAction.

Request Body

Any combination of: title, core_competence, basic_competence, indicator, learning_objective, main_material, learning_method, media_tools, learning_source, learning_activities, assessment

```
{
  "success": true,
  "message": "RPP berhasil disimpan.",
  "updated_fields": [
    "learning_objective",
    "main_material"
  ]
}
```

5.6 Regenerate Field

POST /api/lesson-plans/{id}/regen/{field}

Regenerate a single RPP field using AI. Max 2 regenerations per field. Complex fields (learning_activities, assessment) use Sonnet; others use Haiku.

```
{
  "success": true,
  "data": {
    "field": "learning_activities",
    "value": "<h3>Pendahuluan (15 menit)</h3><p>...</p>",
    "regen_remaining": 1,
    "model_used": "claude-sonnet-4-5-20250929"
  }
}
```

5.7 Regen Limits

GET /api/lesson-plans/{id}/regen-limits

```
{
  "success": true,
  "data": {
    "core_competence": {
      "used": 0,
      "max": 2,
      "remaining": 2
    },
    "basic_competence": {
      "used": 1,
      "max": 2,

```

```

    "remaining": 1
  },
  "indicator": {
    "used": 2,
    "max": 2,
    "remaining": 0
  },
  "...": "all 10 fields"
}

```

5.8 Submit for Review

POST /api/lesson-plans/{id}/submit

Submit RPP for admin review. Only works for draft/revision status. Uses SubmitLessonPlanAction which notifies all admin users.

```

{
  "success": true,
  "message": "RPP berhasil disubmit untuk review."
}

```

5.9 Export to DOCX

GET /api/lesson-plans/{id}/export

Download approved RPP as a DOCX file. Only works for approved status. Increments export_count. Returns binary file with Content-Type: application/vnd.openxmlformats-officedocument.wordprocessingml.document

5.10 List Plans

GET /api/lesson-plans

Parameter	Type	Required	Description
teacher_id	uuid	No	Filter by teacher
school_id	uuid	No	Filter by school
status	string	No	draft submitted approved revision

Parameter	Type	Required	Description
subject_id	uuid	No	Filter by subject
class_id	uuid	No	Filter by class
ai_generated	boolean	No	Filter AI-generated only
per_page	integer	No	Items per page (1-50, default 15)

5.11 Pending Review (Admin)

GET /api/lesson-plans/review/pending

Parameter	Type	Required	Description
school_id	uuid	Yes	School UUID
per_page	integer	No	Items per page (1-50, default 15)

5.12 Approve RPP (Admin)

POST /api/lesson-plans/{id}/approve

Approve a submitted RPP. Uses ApproveLessonPlanAction which sets timestamps and notifies the teacher.

Parameter	Type	Required	Description
approved_by	uuid	Yes	Approver user UUID
note_admin	string	No	Optional admin note (max 2000 chars)

```
{
  "success": true,
  "message": "RPP berhasil disetujui."
}
```

5.13 Request Revision (Admin)

POST /api/lesson-plans/{id}/revision

Return RPP for teacher edits. Uses RequestLessonPlanRevisionAction which notifies the teacher.

Parameter	Type	Required	Description
note_admin	string	Yes	Revision note (max 2000 chars)

```
{
  "success": true,
  "message": "RPP dikembalikan untuk revisi."
}
```

6. Frontend Integration Flows

6.1 Recommendation Dashboard

Step 1: Load class summary for dashboard overview

```
GET /api/recommendations/class/{class_id}/summary
```

Step 2: Display filtered recommendation list

```
GET /api/recommendations?teacher_id={uuid}&class_id={uuid}&status=pending
```

Step 3: Load detail when teacher clicks a recommendation

```
GET /api/recommendations/{id}
```

Step 4: Update status as teacher progresses

```
PATCH /api/recommendations/{id}/status { "status": "in_progress" }
```

Step 5: Generate new recommendations (handle 202 for async, 429 for rate limit)

```
POST /api/recommendations/generate { teacher_id, class_id, subject_id }
```

6.2 Teaching Material Flow

Step 1: Check cache before generating

```
GET /api/generated-materials/check-cache?teacher_id={uuid}&chapter_id={uuid}
```

Step 2a: If cached, load directly

```
GET /api/generated-materials/{material_id}
```

Step 2b: If not cached, generate

```
POST /api/generated-materials/generate { teacher_id, subject_id, chapter_id }
```

Step 3: Display material content, quiz, and references in tabs

Step 4: Regenerate quiz (adds) or references (replaces) using regen_status counts

```
POST /api/generated-materials/{id}/regenerate-quiz
POST /api/generated-materials/{id}/regenerate-reference
```

6.3 RPP Generation & Approval

Teacher Flow

1. Generate RPP with required parameters
2. Display editor with all 10 fields and regen limit badges
3. Auto-save fields on edit (debounce 500ms recommended)
4. Regenerate specific fields as needed (check remaining count)
5. Submit for admin review when ready

Admin Flow

1. View pending RPPs for their school
2. Review RPP detail with all content fields
3. Approve or request revision with admin notes
4. After approval, teacher can export to DOCX

6.4 Async Polling

When backend operates in async mode, all generate endpoints return 202:

```
// 1. Call generate endpoint
const res = await fetch("/api/lesson-plans/generate", {
  method: "POST",
  headers: { Authorization: "Bearer " + token, "Content-Type": "application/json" },
  body: JSON.stringify(payload)
});

// 2. Check response code
if (res.status === 202) {
  const { job_id } = await res.json();
  showSpinner();
  const result = await pollJob(job_id); // See Section 2.4
  hideSpinner();
  displayResult(result);
} else {
```

```
const { data } = await res.json();
displayResult(data); // Sync mode
}
```

6.5 Error Handling

Code	Scenario	Frontend Action
422	Validation error	Show error message, highlight invalid fields
429	Rate limit exceeded	Show limit message, disable generate button temporarily
404	Resource not found	Show "not found", redirect to list
401	Token expired	Redirect to login, clear stored token
500	Server error	Show generic error, suggest retry

6.6 Endpoint Summary

Method	Endpoint	Description
GET	/api/ai-jobs	List AI jobs
GET	/api/ai-jobs/{id}	Get job status & result
POST	/api/recommendations/generate	Generate for class
POST	/api/recommendations/generate-student	Generate for student
GET	/api/recommendations	List recommendations
GET	/api/recommendations/{id}	Show recommendation
PATCH	/api/recommendations/{id}/status	Update status
GET	/api/recommendations/class/{id}/summary	Class summary
POST	/api/generated-materials/generate	Generate material
GET	/api/generated-materials/check-cache	Check cache
GET	/api/generated-materials	List materials
GET	/api/generated-materials/{id}	Show material
POST	/api/generated-materials/{id}/regenerate-quiz	Regen quiz (adds)
POST	/api/generated-materials/{id}/regenerate-reference	Regen refs (replaces)
POST	/api/lesson-plans/generate	Generate RPP
GET	/api/lesson-plans	List plans
GET	/api/lesson-plans/review/pending	Pending review

Method	Endpoint	Description
GET	/api/lesson-plans/{id}	Show RPP
PATCH	/api/lesson-plans/{id}	Auto-save fields
POST	/api/lesson-plans/{id}/regen/{field}	Regen field
GET	/api/lesson-plans/{id}/regen-limits	Regen limits
POST	/api/lesson-plans/{id}/submit	Submit for review
GET	/api/lesson-plans/{id}/export	Export DOCX
POST	/api/lesson-plans/{id}/approve	Approve (admin)
POST	/api/lesson-plans/{id}/revision	Request revision